

COMPUTER EXERCISE 3

VALENCIA 2019

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1. EXERCISE 1

Run the code given in the exercise and the lmer function

```
> y <- c(13.61728, 13.61728, 13.42423, 13.01030, 13.97940, 13.42423, 14.31364,
+ 13.97940, 13.42423, 13.42423, 13.61728, 13.61728, 14.31364, 12.78754,
+ 13.61728, 12.55273, 13.97940, 15.18514, 14.31364, 14.31364, 14.77121,
+ 14.47158, 13.80211, 13.42423, 14.14973, 14.77121, 14.77121, 14.62398,
+ 14.62398, 15.68202, 13.80211, 14.47158, 15.79784, 15.79784, 16.12784,
+ 15.79784, 15.79784, 15.05150, 15.79784, 16.23249, 15.44068, 16.02060,
+ 16.12784, 15.68202, 14.47158, 15.56303, 14.77121, 15.68202, 11.46128,
+ 12.04120, 11.76091, 12.78754, 13.01030, 13.42423, 12.04120, 12.78754,
+ 12.30449, 12.55273, 13.01030, 12.55273, 10.79181, 12.30449, 11.76091,
+ 11.13943, 16.62758, 15.56303, 14.77121, 14.62398, 15.56303, 14.91362,
+ 14.77121, 15.05150, 15.05150, 14.91362, 14.77121, 15.05150, 13.97940,
+ 14.62398, 14.91362, 15.05150)
> id <- c( 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
+ 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
+ 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3,
+ 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4,
+ 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5 )
> #Collect all the data in a data frame
> estrone.data <- data.frame(y=y, id=as.factor(id))
> library(lme4)
> mod1 <- lmer(y~1+(1|id), data=estrone.data)
> summary(mod1)

Linear mixed model fit by REML ['lmerMod']
Formula: y ~ 1 + (1 | id)
Data: estrone.data
```

REML criterion at convergence: 157.8

Scaled residuals:

Date: May 20, 2019.

	Min	1Q	Median	3Q	Max
	-2.56578	-0.41059	0.08074	0.57328	2.84351

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	1.7494	1.3227
Residual		0.3254	0.5705

Number of obs: 80, groups: id, 5

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	14.1751	0.5949	23.83

The residual variance is 0.3254 and the random effect variance is 1.7494.

2. EXERCISE 2

The hglm2 results and diagnostic plots are given with the following code

```
> library(hglm)
> mod1.hglm <- hglm2(y~1+(1/id), data=estrone.data)
> summary(mod1.hglm)

Call:
hglm2.formula(meanmodel = y ~ 1 + (1 | id), data = estrone.data)
```

```
-----
MEAN MODEL
-----
```

Summary of the fixed effects estimates:

	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	14.1751	0.5946	23.84	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note: P-values are based on 75 degrees of freedom

Summary of the random effects estimates:

	Estimate	Std. Error
(Intercept) id:1	-0.6229	0.6046
(Intercept) id:2	0.2690	0.6046
(Intercept) id:3	1.4431	0.6046

```
(Intercept)| id:4  -1.9196    0.6046
(Intercept)| id:5   0.8303    0.6046
```

```
-----
DISPERSION MODEL
-----
```

NOTE: h-likelihood estimates through EQL can be biased.

Dispersion parameter for the mean model:
[1] 0.3254369

Model estimates for the dispersion term:

Link = log

Effects:

Estimate	Std. Error
-1.1226	0.1632

Dispersion = 1 is used in Gamma model on deviances to calculate the standard error

Dispersion parameter for the random effects:
[1] 1.749

Dispersion model for the random effects:

Link = log

Effects:

. Random1	
Estimate	Std. Error
0.5593	0.7112

Dispersion = 1 is used in Gamma model on deviances to calculate the standard error

EQL estimation converged in 3 iterations.

```
> plot(mod1.hglm)
```

3. EXERCISE 3

The easiest way to get the design matrices is to use the `model.matrix()` function.

```
> X <- as.matrix(model.matrix(y~1, data=estrone.data))
> Z <- as.matrix(model.matrix(y~0+id, data=estrone.data))
> mod11.hglm <- hglm(y=y, X=X, Z=Z)
> summary(mod11.hglm)
```

Call:

```
hglm.default(X = X, y = y, Z = Z)
```

```
-----
MEAN MODEL
-----
```

Summary of the fixed effects estimates:

	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	14.1751	0.5946	23.84	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note: P-values are based on 75 degrees of freedom

Summary of the random effects estimates:

	Estimate	Std. Error
id1	-0.6229	0.6046
id2	0.2690	0.6046
id3	1.4431	0.6046
id4	-1.9196	0.6046
id5	0.8303	0.6046

```
-----
DISPERSION MODEL
-----
```

NOTE: h-likelihood estimates through EQL can be biased.

Dispersion parameter for the mean model:

```
[1] 0.3254369
```

Model estimates for the dispersion term:

```
Link = log
```

```
Effects:
```

Estimate	Std. Error
-1.1226	0.1632

Dispersion = 1 is used in Gamma model on deviances to calculate the standard error

Dispersion parameter for the random effects:

```
[1] 1.749
```

Dispersion model for the random effects:

```
Link = log
```

```
Effects:
```

```
.|Random1
  Estimate Std. Error
    0.5593    0.7112
```

Dispersion = 1 is used in Gamma model on deviances to calculate the standard error

EQL estimation converged in 3 iterations.

```
> plot(mod11.hglm)
```

4. EXERCISE 4

95 % CI can be computed as $\exp(0.5593 \pm 1.96 \cdot 0.7112)$, which gives the interval $[0.43, 7.05]$.

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